

RESEARCH ARTICLE

Behavioral Patterns in Micro-lending: Enhancing Credit Risk Assessment with Collaborative Filtering and Federated Learning

Asma Aldrees¹ · Sana Shahab² · Ashit Kumar Dutta³ · Waseem Ahmad⁴ · Mohd Anjum⁵

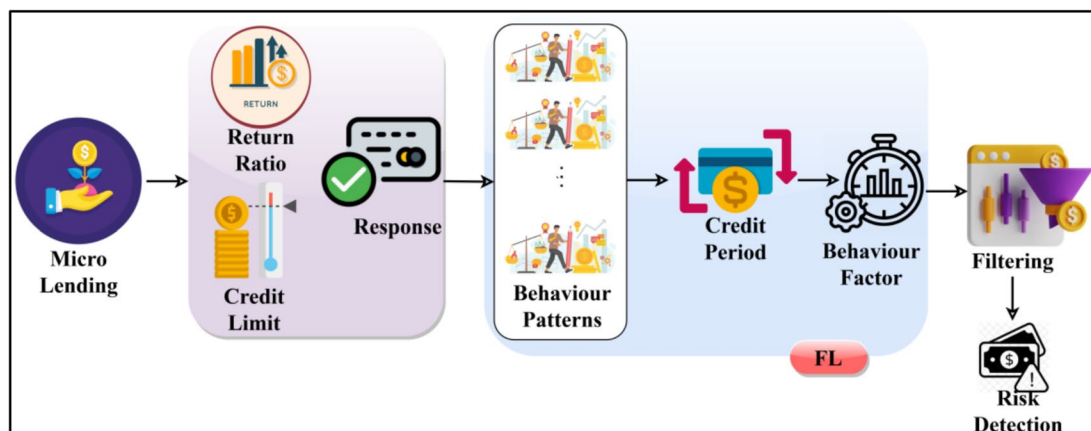
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Abstract

Credit risk assessment uses finance-based behavioural patterns for micro-lending purposes and organizations. The repayment behaviour and credit stability patterns are analyzed across varying repayment tenures and financed amounts. Due to limited borrower data and fluctuating financial patterns, micro-lending platforms have substantial hurdles when it comes to effectively evaluating credit risk. This article introduces a Collaborative Filtering Method using Lending Pattern Analysis (CFM-LPA). The proposed method is enhanced through collaborative federated learning, enabling the analysis of these patterns. This approach evaluates the return rate, credit limit, and consumer response behaviours. Federated learning processes one or more of these factors to assess diverse lending patterns. Based on these evaluations, the behavioural factor is updated for each return period, influencing the credit risk for subsequent return periods and supporting the financial stability of micro-lending operations. The model is trained individually on the identified factors, allowing the behavioural factor to be filtered. New credit risks are identified using this filtered factor from the previous return period. These insights help define new behavioural patterns for the specified credit limit. The proposed method enhances risk detection accuracy by 14.03% and improves return rate analysis by 13.28% across financed amounts. The above abstract is also graphically presented.

Graphical Abstract



Keywords Behavior pattern · Collaborative filtering · Credit risk · Federated learning · Micro-lending



1 Introduction

Credit risk assessment is a must in every micro-lending organization. The actual credit risks are assessed using finance-based behavioural patterns in micro-lending organizations. The behavioural patterns provide stability, interest rate, and return capacity [1]. A unique credit risk assessment strategy is used in micro-lending organizations. The assessment strategy uses logistics functions to determine the financial risks that may occur after lending small funds or loans [2, 3]. The exact severity level of the risks is calculated to produce feasible information for assessment services. The strategy predicts the successful finding of factors minimizing financial crises or risk rates in money lending services [4]. A credit risk analysis technique is also used to analyze the risk ratio for micro-lending services. The analysis technique evaluates the potential risk factors that cause credit risks to the services [5]. The technique identifies the interest rate and predicts the credit risks based on financial stabilities. The analysis technique provides the possibility of risks, which reduces the computational complexity of lending. The technique improves the performance and efficiency of assessment services [6].

Behavioural patterns are analyzed and evaluated, which produces an effective dataset for credit risk prediction in lending applications. The patterns provide details such as financial status, credit details, interest rates, and the impact of micro-lending on customers [7]. A behavioural pattern analysis approach predicts credit risks in micro-lending services. The analysis approach evaluates the impact of loan borrowers' financial status and consistency while paying the debts [8]. The approach identifies the difference between the behavioural patterns and produces feasible data for detection processes. The analysis approach elevates the accuracy rate of credit risk detection and enhances the efficiency range of the lending systems. A tracing detection method is used in lending systems for credit risk prediction [9, 10]. The detection method uses software to analyze the behavioural pattern of the borrower and lender's activity on online platforms. The method provides empirical features to calculate the credit risks as per severity. The method also reduces the computational cost during detection, enhancing the systems' performance. The detection method minimizes the latency range and provides optimal user lending services [11, 12].

Machine learning (ML) methods are used in micro-lending systems for credit risk analysis services. ML methods improve the detection services' precision and accuracy [13]. A credit risk assessment using an ML algorithm is used in micro- and macro-lending applications. The assessment model addresses the common issues that cause application credit risks [14]. The model uses a linear learning analysis method to analyze the overheads and factors of the systems. The assessment model also investigates the actual cause of risks using ML, which minimizes the energy consumption range of the systems [15, 16]. The assessment model maximizes the credit risk prediction and reduces the financial risk factors while lending money to borrowers [17]. A random forest algorithm-based credit risk analysis technique is also used in lending applications. The random forest algorithm analyses the dynamic and systemic credit features to evaluate the actual stages of risks [18]. The analysis technique provides a feasible solution to solve the issues from the applications. It also provides companion strategies to reduce risk rates from the systems. The algorithm reduces micro-lending services' fault and credit risk ratio [19]. The role of consumer behaviour decides the credit risk and further lending in small-scale financial services. This customer/consumer behaviour is defined by various factors that influence credit return policies. A precise credit level/ risk level decision will be made based on the fluctuating influencing factors. Motivated by this factor, the proposed credit risk analysis method is designed and discussed.

1.1 Research Gap

Data shortage, borrower variety, and privacy issues contribute to micro-lending platforms' ongoing struggles with proper credit risk assessment. Due to security concerns and a lack of flexibility for changing financial patterns, traditional credit scoring algorithms often depend on centralized data collecting. Moreover, micro-lending situations often do not have access to the large amounts of labelled data needed by current machine learning-based

methods, resulting in less-than-ideal risk projections. However, this research is not fit for decentralized financial ecosystems since they do not adequately include privacy-preserving approaches in behavioural analysis for credit evaluation. Accurate credit risk assessment must be achieved in micro-lending situations while protecting user data's privacy and security.

1.2 Objectives

This research focuses on the Collaborative Filtering Method utilizing Lending Pattern Analysis (CFM-LPA), an attempt to improve micro-lending credit risk assessment via the integration of federated learning and collaborative filtering. The main goal is to guarantee data privacy via decentralized learning and increase the accuracy of credit risk prediction by using borrower behaviour patterns. A scalable and adaptable model that can capture shifting financial habits is also one of the article's goals, which aims to improve lending practices. More informed lending choices in micro-lending contexts may be supported by this research's robust, privacy-preserving, and data-driven methodology, which addresses the constraints of existing credit scoring methods.

The article makes the following contributions:

1. Introducing and discussing a novel credit risk analysis method supported by federated learning and collaborative learning techniques.
2. Exploring the role of various factors influencing financial stability provides insights into their combined impact through comprehensive analysis.
3. A data-driven approach can be defined to verify the self-analytical factors of the proposed method.
4. A comparative analysis will be presented to evaluate and validate the efficiency of the proposed method.

The rest of the paper is prearranged as follows: Sect. 2 presents the related works, section suggests the CFM-LPA model, Sect. 4 deliberates the results and discussion and Sect. 5 concludes the research paper.

2 Related Works

Zhuang et al. [20] created the GAN-LightGBM model to analyze credit risk in financial technology. The method employs Generative Adversarial Networks (GAN) to address data imbalance and a Light Gradient Boosting Machine (LightGBM) to estimate credit risk. The strategy improves model performance by creating realistic synthetic data and increasing forecast accuracy. The approach provides extensive credit risk assessment capabilities for financial applications. Aruleba et al. [21] used ensemble classifiers using SMOTE-ENN. The approach combines Random Forest, Adaptive Boosting, and XGBoost classifiers with Synthetic Minority Over-sampling Technique-Edited Nearest Neighbour (SMOTE-ENN) to increase model interpretability and alleviate class imbalance. The method incorporates Shapley Additive Explanations for feature importance analysis. The strategy improves model transparency and ensures a reliable credit risk assessment.

Yang et al. [22] presented a multi-stage ensemble model. The technique extracts relevant aspects by combining behavioural and non-financial data. To solve class imbalance, the method uses bagging-based oversampling. The strategy increases small and medium enterprises' credit risk prediction and management tactics. Gamba-Santamaria et al. [23] used intrinsic estimators with penalized regression. The technique separates loan risk into payment capacity and bank risk components. The strategy addresses multicollinearity and results in more accurate credit risk analysis. The strategy helps policymakers comprehend systemic financial risks better, which provides deeper insights.

Haitao et al. [24] designed an integrated supply chain management system. The strategy improves real-time data sharing between suppliers, dealers, and customers, which leads to better credit risk management. The technique boosts operational efficiency and decision-making capabilities. The strategy promotes better risk management

on e-commerce platforms. Wang et al. [25] used Nonlinear Least Squares Support Vector Machine to develop a specialized index system for credit risk classification in online supply chains. The approach improves analytical precision and overall generalization performance. The strategy outperforms existing credit risk models in terms of risk classification. The method improves online credit risk evaluation processes.

Rao et al. [26] used Particle Swarm Optimisation to fine-tune eXtreme Gradient Boosting hyperparameters to improve auto loan credit risk assessment. The strategy optimizes parameter choices to increase model accuracy and efficiency. The strategy lowers computational expenses while improving prediction performance. The strategy greatly improves auto loan credit risk prediction.

Shetabi [27] suggested an evolving ensemble feature technique. The technique uses numerous algorithms to assess credit risks dynamically. The technique improves predictive accuracy and computational efficiency. The technique adjusts to changing risk factors in the financial technology industry. Xia et al. [28] used ML algorithms for classification and risk prediction to manage risk on online loan platforms. The technique examines platform activity to identify harmful behaviours and execute preventive actions. The method enhances the security and management of online lending systems. The strategy improves risk identification and mitigation strategies.

Sinkey Jr. et al. [29] statistically analyzed risk factors. The technique examines the factors influencing loan losses and banks' risk management practices. The approach helps to understand the economic implications of lending behaviour. The technique offers useful information for enhancing financial strategies in the banking industry. Li et al. [30] focused on loan profit forecast. The method combines financial and non-financial information to provide a thorough risk analysis. The strategy improves prediction accuracy and enables risk management for small and medium enterprises loans. The strategy offers actionable insights on reducing small business credit risks.

Wang et al. [31] used ensemble learning approaches to enhance their forecasts for student loan default. The approach uses many classifiers to increase prediction resilience and accuracy. The method corrects the class imbalance and improves prediction performance. The strategy provides insight into educational initiatives and improves loan default risk management. Zhang et al. [32] created a visual early warning system to detect credit risks associated with college net loans. The approach analyses several danger indicators in real-time to generate early warnings. The strategy eliminates dangerous loans by timely interventions. The strategy improves debt management and student safety at educational institutions.

Liu et al. [33] investigated risk management solutions for bank mortgage loans to improve home loan risk management. The strategy improves decision-making by combining financial analysis and prediction models. The strategy improves home loan risk management by identifying significant risk elements. The method establishes a reliable framework for evaluating mortgage loan risks. Carannante et al. [34] proposed a climate risk-sharing approach. The technique uses climatic risk indicators to determine loan viability. The strategy decreases exposure to climate-related risks in the tourism business. The strategy promotes strong financial systems for both borrowers and lenders. Liu et al. [35] investigated loan securitizations and risk-sharing systems to reduce revenue volatility for small enterprises. The strategy incorporates income insurance to reduce income volatility. The strategy improves financial stability in small business-dependent areas. The strategy promotes total financial security in the small business sector. Table 1 shows the summary of existing models.

Micro-lending services rely on heterogeneous patterns classified under services and consumer behaviours. The conventional methods rely on return rate and customer history for further lending and validation. ML-based solutions rely on the statistical computation of return rate to increase the credit level and thereby infer the reduction rates. However, this is feasible for a large financial service, whereas in a low-interest-rated financial service, the different patterns (customer return and lending) impact credit risks differently. These include delayed response or failed returns across increasing scheduled tenures. This article proposes a pattern-converted behaviour factor assessment method using federated learning to address these diverse issues. This method validates financial and customer factors to identify filtered risks in lending credits. Traditional machine learning models and ensemble approaches comprise the bulk of the existing literature on micro-lending credit risk assessment; however, these methods often depend on centralized data sources, which might lead to privacy problems. A lack of complete

Table 1 Summary of the existing models

Author(s)	Method	Results	Advantage	Limitations
Zhuang et al. [20]	GAN-LightGBM (GAN for data balancing + LightGBM for risk estimation)	Improves credit risk prediction accuracy	Generates realistic synthetic data, enhances prediction reliability	GAN-generated data may not always capture real-world complexities
Aruleba et al. [21]	Ensemble classifiers (RF, AdaBoost, XGBoost) + SMOTE-ENN	Increases model interpretability and alleviates class imbalance	Incorporates SHAP for feature importance analysis, ensures transparency	Computationally expensive due to multiple ensemble models
Yang et al. [22]	Multi-stage ensemble model + bagging-based oversampling	Enhances SME credit risk management	Uses behavioral and non-financial data for better assessment	May not generalize well to different financial ecosystems
Gamba-Santamaria et al. [23]	Intrinsic estimators + penalized regression	Separates loan risk into payment capacity and bank risk	Addresses multicollinearity, improves systemic risk comprehension	Requires extensive financial datasets for precise estimations
Haitao et al. [24]	Integrated supply chain management system	Enhances real-time data sharing for credit risk management	Improves decision-making and operational efficiency	Requires high integration among financial institutions and supply chains
Wang et al. [25]	Nonlinear Least Squares Support Vector Machine	Improves credit risk classification in online supply chains	Enhances analytical precision and generalization	Computationally intensive and sensitive to hyperparameter selection
Rao et al. [26]	PSO-optimized XGBoost	Enhances auto loan credit risk prediction	Reduces computational cost, optimizes hyperparameters	Limited effectiveness in highly volatile economic environments
Shetabi [27]	Evolving ensemble feature selection	Dynamically adjusts credit risk models	Improves predictive accuracy and computational efficiency	Requires frequent updates to adapt to changing risk factors
Xia et al. [28]	ML classification models for online lending risk	Identifies harmful behaviors in online loans	Enhances security and risk mitigation strategies	May struggle with adversarial manipulation of loan platforms
Sinkey Jr. et al. [29]	Statistical analysis of risk factors	Improves understanding of loan losses and risk practices	Provides economic insights into banking risk strategies	May not capture emerging financial risks in digital lending
Li et al. [30]	Loan profit forecasting (financial + non-financial data)	Enhances risk analysis for SME loans	Improves prediction accuracy and credit risk management	Performance depends on the availability of diverse financial data
Wang et al. [31]	Ensemble learning for student loan default prediction	Improves resilience and accuracy of predictions	Addresses class imbalance, enhances risk management	May have limited interpretability for policymakers
Zhang et al. [32]	Visual early warning system for student loan risks	Provides real-time credit risk warnings	Enables proactive debt management	Effectiveness depends on the accuracy of warning indicators
Liu et al. [33]	Bank mortgage risk management models	Enhances home loan risk assessment	Identifies key risk elements for better decision-making	Requires extensive historical mortgage data for training
Carannante et al. [34]	Climate risk-sharing approach	Reduces exposure to climate-related financial risks	Strengthens financial systems for borrowers and lenders	May not fully mitigate climate-induced economic shocks
Liu et al. [35]	Loan securitization + risk-sharing for SMEs	Improves financial stability for small businesses	Reduces income volatility and enhances security	Implementation complexity and regulatory challenges

integration of borrower behavioural patterns limits the forecasting accuracy of techniques like GAN-LightGBM and SMOTE-ENN, which address data imbalance and feature selection. In addition, the difficulties of data fragmentation among financial institutions are mostly ignored in most research, resulting in inadequate risk evaluations and incomplete borrower profiles. Research on the combination of federated learning with collaborative filtering for customized credit risk prediction is still in its early stages, but it provides a privacy-preserving option. Furthermore, there are significant inequalities in accountability and long-term dependability caused by the fact that current models seldom ever deal with problems such as model drift, adversarial robustness, and demographic biases in loan choices. Improving risk detection while guaranteeing data privacy and transparency is the goal of the proposed CFM-LPA paradigm, which employs federated learning and collaborative filtering to fill these gaps.

3 Data Description

Credit risk analysis data from [36] is used to validate the proposed method using fuzzy optimization. The data is classified under 12 features utilizing personal and loan-related information. The status and history of the data provided are used to analyze the behaviour factor. The behaviour factor computation process uses the return rate, prompt return, and credit limit. Thus, the data used to evaluate the proposed method is described in Fig. 1.

The data are split into three blocks: initial input (return + Response) and behaviour pattern. For this purpose the above-referenced data with the following feature values is used: the age ranges between (20 and 45), income (4 K to 2039 K), ownership (rented, own, mortgage, employment tenure (1 to 31 yrs), loan purpose (personal, education, venture, medical, home needs), loan grade (A to D), loan amount (0.5 K to 35 K), interest rate (5.42 to 23.22), loan status (0-nil, 1-pending) per cent of loan income (0 to 0.83), the credit history (2 to 30 years). The above data are provided for 32,581 record sets as collected from the dataset. The notable parameters: credit history, grade, status, and income % decide the behaviour patterns of the consumers. This pattern detection finalizes the credit risk under different lending periods (Refer to Fig. 1).

4 Proposed Collaborative Filtering Method Using Filtering Pattern for Credit Risk Analysis

Credit risk assessment is one of the most important aspects of the sustainability and profitability of micro-lending organizations. In micro-lending, understanding the financial behaviour of borrowers is critical to minimize defaults and optimize credit distribution. This incorporates a Collaborative Filtering Method (CFM) to enhance credit risk assessment using Lending Pattern (LP) analysis. The critical factors, including return

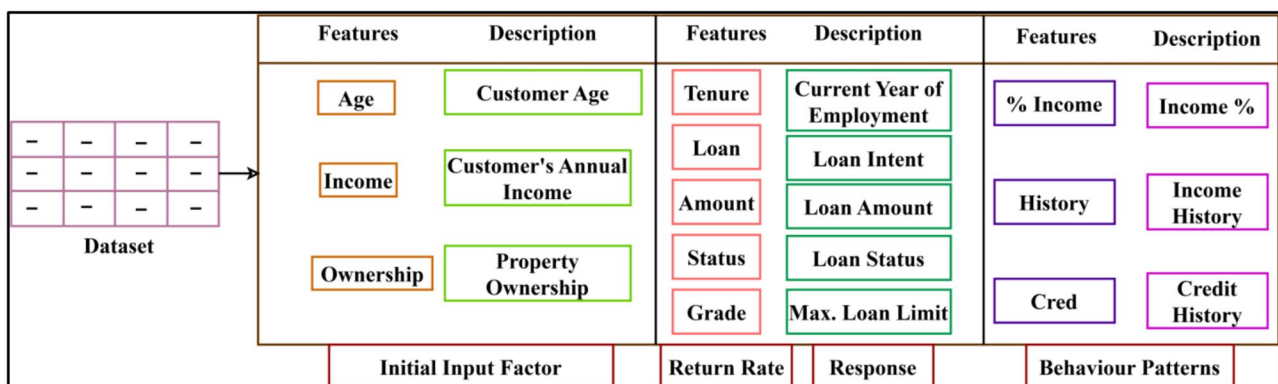


Fig. 1 Data used for evaluation

rates, credit limits, and consumer response behaviours, were evaluated through collaborative federated learning. The federated approach allows for the decentralized analysis of lending patterns in a manner that ensures data privacy and security requirements in any financial system. The CFM analyzes such factors over successive repayment periods, identifies behaviour trends, and refines its credit risk assessment. This approach offers strong credit stability prediction and limit optimization to ensure the financial stability of micro-lending organizations. The following equation $M_L(t)$ measures the micro-lending factors that help to monitor and predict the risk in the environment at a time.

$$M_L(t) = (C_{lm} + Br_C \times ((I_{rate} \times E_{con}) + B_{fact}) + C_{risk}) \times (1 + B_{fact}(t + 1)). \quad (1)$$

To predict the risk in micro-lending, it is important to monitor and analyze factors that lead to environmental risk. The credit limit is denoted as C_{lm} which is provided to the borrower with a particular limit. It helps to measure how the credit is extended to the borrower over time. A high credit limit may indicate a financial risk and affect the behavioural pattern. The borrower's credit score is represented as Br_C and B_{fact} measures the behaviour factor of the borrower. It helps to identify the external risk factors. The term $(I_{rate} \times E_{con})$ measures the combined analysis of interest rates and economic conditions such as inflation with variations in credit rate. The credit risk is denoted as C_{risk} which monitors the risk factors during the credit period. The term $B_{fact}(t + 1)$ measures the behaviour factor for the next time to predict the upcoming environmental credit risk. This helps monitor the micro-lending organization to adapt to changing financial conditions and mitigate financial risk. The return rate is measured as R_{rate} in the equation below.

$$\left. \begin{aligned} R_{rate} &= \left(\frac{R_{amt-tot}}{amt} \right) + \exp(C_{lm} \times Br_C) + M_L(t) \\ \text{where } R_{rate} &> C_{lm} \forall (amt - E_{con}) \\ R_{amt-tot} &= \sum_{i=1}^t \left(B_{fact,i} \times \left(\frac{C_{risk} \times I_{rate,i}}{1 - (1 + I_{rate,i})^t} \right) \right) \end{aligned} \right\}. \quad (2)$$

Here, the total return amount is denoted as $R_{amt-tot}$ and the total amount considered for micro-lending is represented as amt . The term $\left(\frac{R_{amt-tot}}{amt} \right)$ computes the ratio of the amount to be returned by the borrower over the amount they borrowed. It is evaluated by analyzing the borrower's credit limit and behaviour factor. When $R_{rate} > C_{lm} \forall (amt - E_{con})$ indicates that the return rate exceeds the credit limit, which ensures a possibility of risk. The total amount is monitored for each month i to t to avoid complications in lending. The term $C_{risk} \times I_{rate,i}$ helps to adjust the credit risk based on $B_{fact,i}$ and $(1 + I_{rate,i})^t$ balance the interest rate over time. This helps to monitor both credit risk and the overall amount needed to be returned by the borrower to ensure an accurate return rate in the micro-lending process. The following equation C_{lm} measures the credit limit.

$$\left. \begin{aligned} C_{lm} &= [B_{fact} \times Br_{risk}] + (1 + C_{hist}) \times \left(\frac{C_{risk}}{E_{con}} \right) \\ 0 &\leq Br_{risk} \leq 1 \\ Br_{risk} &= \sum_{i=1}^t (amt + (Br_C - C_{hist})) \times E_{con} \end{aligned} \right\}. \quad (3)$$

In the above equation, the borrower risk during lending is denoted as Br_{risk} which monitors the risk from the borrower's side. The term $(1 + C_{hist})$ measures the influence of credit history and the term $\left(\frac{C_{risk}}{E_{con}} \right)$ defines the environmental factors that cause credit risk. It evaluates the borrower factor after considering the credit history. If $0 \leq Br_{risk} \leq 1$ is true; it normalizes the borrower risk, avoiding deviations in behaviour patterns. The term $(Br_C - C_{hist})$ measures the variations between behavioural patterns and credit history to evaluate the borrower's payment details. The relation between the return rate and the credit limit is diagrammatically presented in Fig. 2.

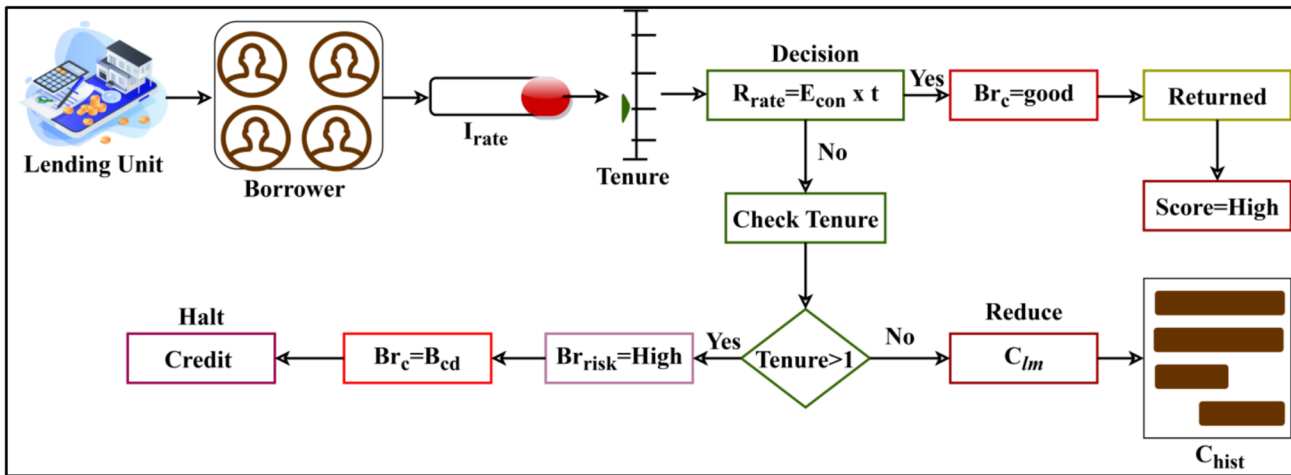


Fig. 2 Illustration of return rate and credit limit relation

The relationship between return rate and C_{lm} is presented using Fig. 2 representations. If $(R_{rate} = E_{con} \times t)$ then $Br_c = \text{good}$, and the rate of return with the score is high. This type of customer exhibits good behaviour patterns without change in repayment. The “loan status = 0” retention in the appropriate tenure reduces the credit risk for the customers. The improper return rate is verified under single/multiple tenures to estimate the high risk/ limit reduction across various (progressive) tenures. The condition $(0 \leq Br_{risk} \leq 1)$ verifies the need for normalization to check if R_{rate} is feasible. This relationship is exploited further to validate the $(Br_c - C_{hist})$ across various tenures (Fig. 2). The suggested CFM-LPA model’s collaborative filtering method uses past lending trends to evaluate borrowers’ similar features and, by extension, their credit risk. Utilizing a combination of user- and item-based filtering algorithms, it generates risk ratings by analyzing lending behaviour, repayment history, and financial transactions. This is made easier using the federated learning architecture, allowing decentralized model training across several banks without exposing sensitive information. This improves generalizability across varied datasets while also preserving privacy. With the use of tools like Federated Averaging (FedAvg), the federated model may improve risk assessment models and reduce data silo problems by aggregating local changes. Further optimization is necessary to address non-IID data distributions, communication costs, and model drift. This will guarantee scalable and resilient performance in real-world financial applications. Based on this factor, the analysis of Br_{risk} is presented in Fig. 3.

Credit risk assessment incorporates risk factors such as default rate, repayment, and credit history. It is monitored with credit limit over the borrowers at a time. The obtained credit risk assessment score is updated using federated learning with LPs from different scenarios. The proposed method ensures high credit risk assessment $C_{risk}(t+1) = (\text{Len}_{\text{adjust}} + \log(Br_c \times C_{hist})) \times (1 + \text{Len}_{\text{pat}})$ which accurately monitors the financial instability. For example, a borrower with a high debt and consistent payments shows high credit risk. A borrower with a stable return rate and timely repayments has a lower credit risk score. This high credit risk helps to identify the borrowers with increasing risks in their tenure (Fig. 3). It measures the organization’s credit policy and risk toward the credit amount. This adjusts the credit limit based on the borrower’s lending capacity to ensure fair lending decisions. The patterns for return and credit stability are analyzed under different repay tenures, and the sum finance which is expressed as $R_{rate}(U_{\text{repay}})$ and $C_{stable}(U_{\text{repay}})$ in the equation below.

$$R_{rate}(U_{\text{repay}}) = \sum_{i=1}^t (B_{\text{fact},i} \times R_{rate}) + (1 - Dr_i) + \left(\frac{Br_c}{\text{amt}} \right) \times C_{lm}, \quad (4)$$

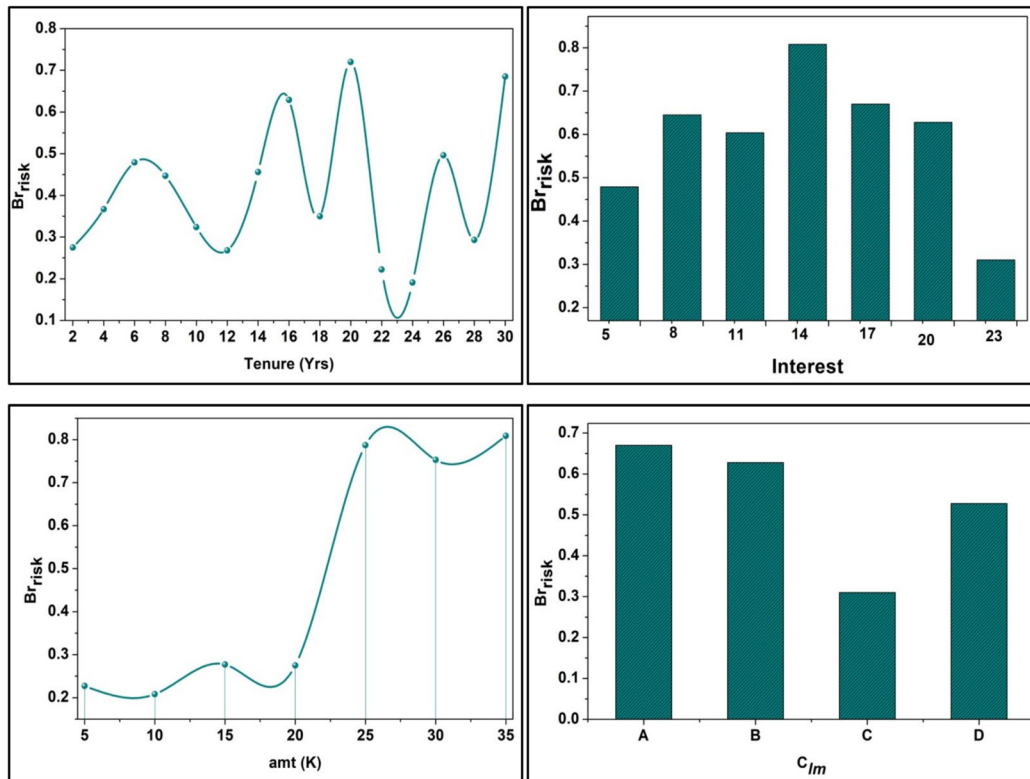


Fig. 3 Credit risk analysis

$$C_{stable}(U_{replay}) = \sum_{i=1}^t (1 + Dr_i) \times \left(\frac{amt \times I_{rate}}{1 + E_{con}} \right) \times \left(\frac{C_{lm} + C_{risk}}{C_{hist}} \right) \times (R_{amt-tot} - Br_C). \quad (5)$$

The different repayment tenure is denoted as U_{replay} in which the term $(B_{fact,i} \times R_{rate})$ measures the behaviour factor with the return rate for the period i to monitor the accurate repayment amount. The default payment by the borrower for each month is denoted as Dr_i in which the term $(1 - Dr_i)$ measures the impact of default payment on the overall lending process. The behaviour patterns were adjusted as $0 \leq C_{lm} \leq 1$ for consistent borrower lending over the entire repayment tenure. The financial reliability of the borrower is considered in the credit stability. A high inflation with instability in credit occurs when $E_{con} \geq 0$. The term $\left(\frac{amt \times I_{rate}}{1 + E_{con}} \right)$ estimates the overall amount that needs to be repaid by the borrower. The return rate analyzes how effectively the amount is repaid over the given tenure, and credit stability measures the borrower's financial stability under different repayment conditions. This helps to analyze the variations in repayment in the micro-lending environment. The following equation $H_{res}(t)$ measures the response rate.

$$H_{res}(t) = (R_{rate} + C_{lm}) \times (R_{rate}(U_{replay}) + C_{stable}(U_{replay})) \times (1 - Dr_i) + (1 + E_{con}). \quad (6)$$

The influence of return rate and credit stability measures the borrower's behaviour to the credit response. The term $(R_{rate}(U_{replay}) + C_{stable}(U_{replay}))$ measures the combined relationship of the return rate with the credit rate. It measures how much of the loan amount is repaid over the tenure and how much is credited to the borrower. A high R_{rate} ensure accurate repayment behaviour and a high C_{stable} indicate high potential risk and stable behaviour. The term $(1 - Dr_i)$ measures the impact of default rate in which a high Dr_i reduce the default from the borrower side. A high E_{con} increases the risk of environmental factors that reduce the response in the

lending process. This helps to predict the borrower's loan repayment and performance to be adjusted for the borrower's specific behaviour. The behaviour pattern detection process is presented in Algorithm 1.

Algorithm 1 Behavior Pattern Detection

Read U_{repay} and $Len_{fed}(t)$ to initialize the behaviour pattern

Monitor $E_{con} \forall U_{repay} \rightarrow E_{con} \geq 0$

Analyze $H_{res}(t)$ for $(R_{rate} + C_{lm})$

Compute $Len_{pat} = \sum_{i=1}^t \left(\frac{amt \times C_{risk}}{1 - (1 + I_{rate})} \right) + (1 - Dr_i)$

Adjust Len_{pat} as $Len_{adjust} = (R_{rate} \times C_{lm}) + H_{res}(t)$

Monitor $C_{risk}(t + 1)$

If $C_{risk}(t + 1) > Len_{adjust}$ then initiate $Len_{fed}(t)$

Evaluate $(Len_{pat} \times C_{risk}(t + 1))$

Compute $Len_{fed}(t) \rightarrow \left(\frac{C_{stable}(U_{repay}) + BrC}{(1 + I_{rate})} \right)$

Analyze $Len_{fed}(t) \propto (1 + (Dr_i \times E_{con}))$

Perform $Ft_{bh-in} = U_{repay}(R_{rate} \times C_{stable}) - Dr(t) + H_{res}(t)$

If Ft_{bh-in} monitors U_{repay} then verify amt_{due} and amt_{paid}

Evaluate $U_{repay} = \left(\frac{amt_{due}}{amt_{paid}} \right) \times amt \times (R_{rate} \times C_{lm})$

If $Ft_{bh}(t) \forall U_{repay}$

Compute $Ft_{bh}(t) = (Len_{fed}(t) \times Ft_{bh-in}) \times (1 + E_{con})$ to monitor the variations in the behaviour patterns of borrowers.

A CFM using an LP helps to analyze the credit risk, which is discussed in the following sections.

4.1 Collaborative Filtering

The historical data of multiple borrowers were evaluated using the CFM and LP to analyze patterns of repayment behaviour and predict future loan performance. It monitors repayment history, credit limits, return rates, and borrower behaviour in different segments to analyze the similarities between borrowers. The identified patterns were used to classify and predict new borrowers' credit and repayment defaults. LP analysis mainly identifies the consistent traits of borrowing behaviour, such as timely repayments, the proportion of loans repaid, and financial stability indicators. CFM enhances lending decisions through federated learning without exchanging borrower-specific data directly to maintain privacy. The system filters relevant behaviour factors to predict the borrower's future repayment behaviour and determine the associated credit risk. The borrower's profile was updated every repayment period to change behaviours for accurate credit risk assessments and better lending decisions in successive cycles. The following equation Len_{pat} measures the process of LPs for credit risk.

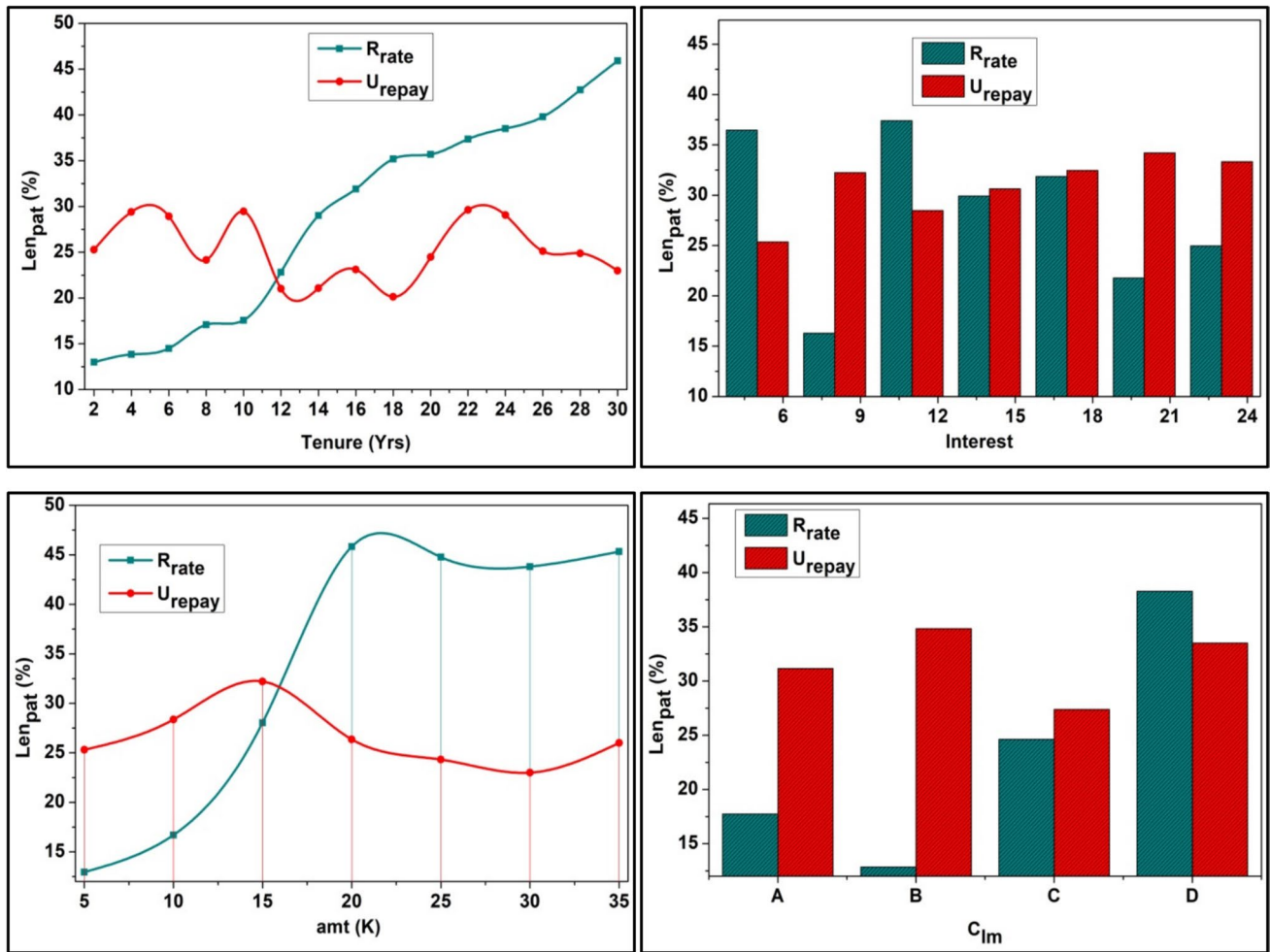


Fig. 4 Lending pattern analysis

$$\left. \begin{aligned} \text{Len}_{\text{pat}} &= \sum_{i=1}^t \left(\frac{\text{amt} \times C_{\text{risk}}}{1 - (1 + I_{\text{rate}})} \right) + (1 - \text{Dr}_i) \\ \text{Len}_{\text{adjust}} &= (R_{\text{rate}} \times C_{\text{lm}}) + H_{\text{res}}(t) \\ C_{\text{risk}}(t+1) &= (\text{Len}_{\text{adjust}} + \log(\text{Br}_C \times C_{\text{hist}})) \times (1 + \text{Len}_{\text{pat}}) + E_{\text{con}} \end{aligned} \right\} \quad (7)$$

In the above equation, the term $\left(\frac{\text{amt} \times C_{\text{risk}}}{1 - (1 + I_{\text{rate}})} \right)$ measures the repayment of individual borrowers using interest rates. This is adjusted using the default rate to avoid fault predictions. The formulation of $\text{Len}_{\text{adjust}}$ evaluates the adjusted LPs to improve the prediction for new borrowers based on the previous response. It ensures privacy during lending between borrowers. The term $C_{\text{risk}}(t+1)$ measures the credit risk for time $(t+1)$ to monitor the next risk period during lending. This combines the adjusted and actual LP with credit history to ensure accurate prediction. Federated learning continuously updates this LP to ensure that the new borrower is assessed based on the current LPs and economic conditions. LP analysis is presented in Fig. 4.

The variations in risk factors may lead to high risk in the micro-lending platform. These variations were monitored and analyzed using pattern analysis. The proposed method, which ensures consistent repayment, ensures high pattern analysis. This high value indicates strong patterns that align with borrowers' financial stability and consistent behaviour. Incorporating federated learning aggregates data across varying lending scenarios with privacy to analyze the patterns. A borrower with a stable pattern of repayments for every month indicates a smooth

lending process without risk. A borrower with irregular repayment and behaviour patterns leads to abnormal patterns and indicates a potential risk during lending. This high pattern analysis enhances the understanding of borrower behaviour over time (Fig. 4). The filtering process is described in Algorithm 2.

Algorithm 2 Filtering Process

```

Initialize  $R_{rate}$  and  $C_{lm}$ 
Read  $C_{risk}$  and  $C_{stable}$ 
Verify  $U_{repay}$  to compute  $Ft_{bh}(t) \forall U_{repay}$ 
Generate  $amt_{due}$  and  $amt_{paid}$ 
Read  $C_{risk}(t+1)$  based on  $H_{res}(t)$ 
Perform  $C_{risk}(t+1) = (Len_{adjust} + \log(Br_C \times C_{hist})) \times (1 + Len_{pat})$ 
Incorporate  $Len_{fed}(t)$ 
Apply  $Len_{fed}(t) \rightarrow (Len_{pat} \times C_{risk}(t+1))$ 
Generate  $U_{repay}$ 
Combine  $U_{repay}(R_{rate} \times C_{stable})$ 
Compute  $U_{repay} = \left(\frac{amt_{due}}{amt_{paid}}\right) \times amt \times (R_{rate} \times C_{lm})$ 
Perform  $C_{period} = (Br_C + C_{hist}) \times \left(1 + \left(\frac{amt}{amt_{mon}}\right)\right)$ 
Generate  $B_{fact}(t)$ 
Evaluate  $C_{period} \rightarrow \max_t C_{period}$  and  $C_{period} \rightarrow \min_t C_{period}$ 
Verify  $B_{fact}(t) = \sum_{i=1}^t \left(C_{period} \times Ft_{bh}(t)\right) \times \left(\frac{o_i}{(1+I_{rate})}\right)$ 
Analyze  $B_{fact}(t) \propto \left(\frac{o_i}{(1+I_{rate})}\right)$ 
Compute  $A_{risk} = \sum_{i=1}^t \left(\frac{C_{risk} \times C_{stable}}{I_{rate} - \max_t C_{period}}\right) - \left(\frac{C_{risk} \times U_{repay}}{I_{rate} - \min_t C_{period}}\right)$ 

```

4.2 Role of Federated Learning

Federated learning is incorporated to learn the pattern variations for multiple lenders and borrowers. In the LP credit risk assessment, federated learning ensures the data privacy of various micro-lending entities. Federated learning evaluates the risk that drives return rates, credit limits, and repayment behaviour among diverse micro-lenders. The system excludes sensitive information related to the borrowers to conform to the LP on data privacy. It generalized patterns in lending to avoid the risk based on high-risk detection. The historical credit rate is constantly updated to capture temporal shifts in borrower behaviour and financial risk. A decentralized method

improves the quality of risk-based lending predictions without affecting borrowers' confidentiality. Federated learning analyzes a minimum of one or all of the above factors to verify different LPs, which is termed as $\text{Len}_{\text{fed}}(t)$ in the equation below.

$$\text{Len}_{\text{fed}}(t) = \sum_{i=1}^t (\text{Len}_{\text{pat}} \times C_{\text{risk}}(t+1)) \times \left(\frac{C_{\text{stable}}(U_{\text{repay}}) + \text{Br}_C}{(1 + I_{\text{rate}})} \right) + (1 + (\text{Dr}_i \times E_{\text{con}})). \quad (8)$$

The actual lending data are processed into an LP for each borrower based on $(\text{Len}_{\text{pat}} \times C_{\text{risk}}(t+1))$ to enhance the lending prediction. The term $\left(\frac{C_{\text{stable}}(U_{\text{repay}}) + \text{Br}_C}{(1 + I_{\text{rate}})} \right)$ monitors the relative importance of the stable repayment credit based on the borrower. It analyzes the environmental conditions that affect the borrower's behaviour. The term $(1 + (\text{Dr}_i \times E_{\text{con}}))$ measures the influence of default rates and environmental conditions that affect the overall LP. This helps monitor the borrower's repayment credit during the entire micro-lending process. It identifies the fluctuations that affect the repayment behaviour in the lending process. The following equation $\text{Ft}_{\text{bh}}(t)$ formulates the behaviour pattern over time.

$$\left. \begin{aligned} \text{Ft}_{\text{bh-in}} &= U_{\text{repay}} (R_{\text{rate}} \times C_{\text{stable}}) - \text{Dr}(t) + H_{\text{res}}(t) \\ U_{\text{repay}} &= \left(\frac{\text{amt}_{\text{due}}}{\text{amt}_{\text{paid}}} \right) \times \text{amt} \times (R_{\text{rate}} \times C_{\text{lm}}) \\ \text{Ft}_{\text{bh}}(t) &= (\text{Len}_{\text{fed}}(t) \times \text{Ft}_{\text{bh-in}}) \times (1 + E_{\text{con}}) \forall U_{\text{repay}} \end{aligned} \right\}. \quad (9)$$

Here, the term $\text{Ft}_{\text{bh-in}}$ measures the initial behaviour pattern to monitor the repayment consistency. It captures the borrower's ability to meet the repayment rate consistently in the lending process. The repayment is measured by evaluating the due amount over the paid amount by the borrower, which is represented as amt_{due} and amt_{paid} . It helps to reduce the behaviour pattern when there is a high default rate in the market. A high U_{repay} indicates the borrower's credit inefficiency, and a low repayment rate indicates a high return rate. The term $(\text{Len}_{\text{fed}}(t) \times \text{Ft}_{\text{bh-in}})$ combines federated lending with the initial behaviour factor to balance the risk factors. The lender can assess the credit risk according to the initial behavioural factor. Financial adaptability captures income fluctuations and spending behaviours; loan purpose categorizes lending patterns (e.g., business vs. personal use) to identify risk profiles; repayment consistency tracks timely and late payments to assess reliability; loan frequency indicates a borrower's dependency on micro-loans; peer influence analyzes how similar borrowers impact creditworthiness; and financial adaptability tracks repayment consistency. By including these parameters in collaborative filtering, the system can estimate credit risk based on previous patterns by mapping borrowers with similar lending habits. Federated learning allows pattern detection among several lenders without sacrificing privacy by allowing decentralized institutions to communicate updated models instead of raw data safely. This $\text{Ft}_{\text{bh}}(t) \forall U_{\text{repay}}$ ensures the model's adaptability to enhance the borrower even during environmental conditions. The following equation C_{period} models the credit period in the lending process.

$$C_{\text{period}} = (\text{Br}_C + C_{\text{hist}}) \times \left(1 + \left(\frac{\text{amt}}{\text{amt}_{\text{mon}}} \right) \right) + \log(1 + I_{\text{rate}}) + \left(\frac{\text{Dr}_i}{E_{\text{con}}} \right), \quad (10)$$

$$f(C_{\text{period}}) = \begin{cases} \text{if } (\text{Br}_C + C_{\text{hist}}) \geq \varphi \text{ and } \text{Dr}_i \leq \varphi \max_t C_{\text{period}} \\ \text{if } (\text{Br}_C + C_{\text{hist}}) < \varphi \text{ and } \text{Dr}_i > \varphi \min_t C_{\text{period}} \end{cases}. \quad (11)$$

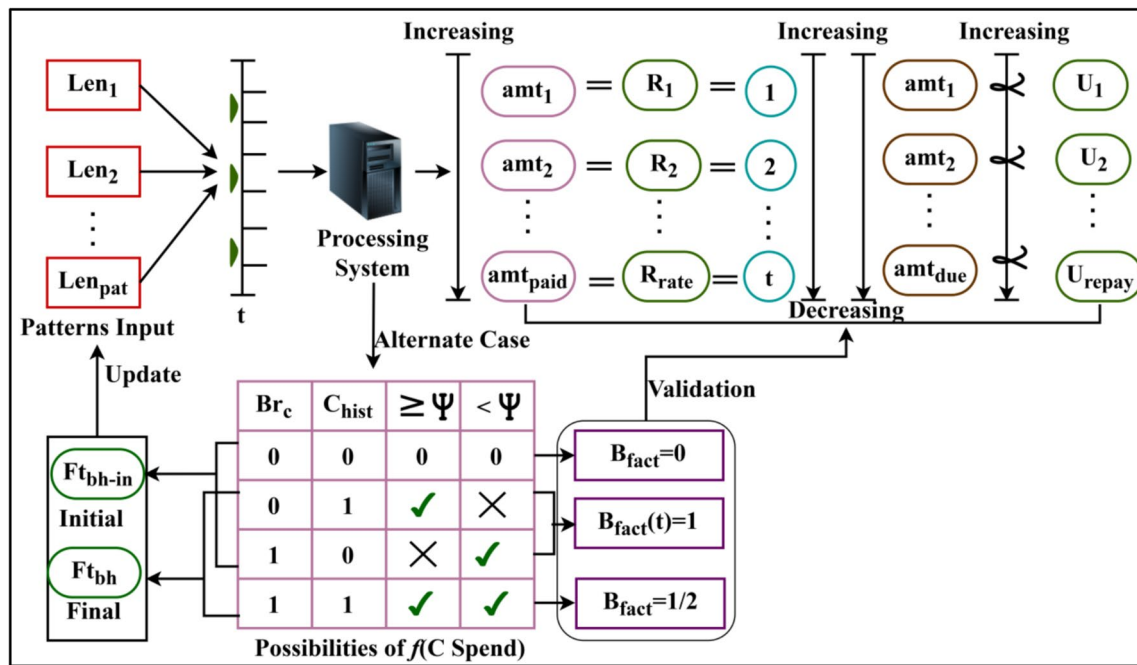


Fig. 5 Behaviour factor estimation using federated learning

The amount to be paid monthly by the borrower is denoted as amt_{mon} which is measured over the total amount that is lent to the borrower. The term $\left(1 + \left(\frac{\text{amt}}{\text{amt}_{\text{mon}}}\right)\right)$ monitors the loan tenure to the borrower and its influence on the risk. The term $(1 + I_{\text{rate}})$ measures the interest needed to repay during the tenure. Borrowers with poor credit and historical scores will have low credit periods. Borrowers with high credit and historical scores will have high credit periods. A high E_{con} reduce the credit period with a high Dr_i should be reduced to minimize the risk. The computation of $f(C_{\text{period}})$ evaluates when the credit period becomes maximum and minimum in the lending process. It is monitored over a predefined threshold φ , which predicts the credit period. When $(Br_c + C_{\text{hist}}) \geq \varphi$ and $Dr_i \leq \varphi$ then the credit period is considered as maximum, and $(Br_c + C_{\text{hist}}) < \varphi$ and $Dr_i > \varphi$ indicates the minimum credit period based on the LP. The following equation $B_{\text{fact}}(t)$ measures the behavior pattern based on the change in the credit period.

$$B_{\text{fact}}(t) = \sum_{i=1}^t (C_{\text{period}} \times Ft_{\text{bh}}(t)) \times \left(\frac{O_i}{(1 + I_{\text{rate}})} \right) \times (Br_c - Dr_i). \quad (12)$$

The term $(C_{\text{period}} \times Ft_{\text{bh}}(t))$ captures the repayment based on the behaviour pattern to ensure consistency. The credit period is adjusted according to $\min_i C_{\text{period}} \leq C_{\text{period}} \leq \max_i C_{\text{period}}$ which helps assess the borrower's repayment. The liability is denoted as O_i which reduces the behaviour factor and indicates the high financial risk in the lending process. The term $\left(\frac{O_i}{(1 + I_{\text{rate}})}\right)$ computes the liability over the influence of interest rate to enhance the behavioural factor. The term $(Br_c - Dr_i)$ evaluates the difference between the borrower credit and default rate to analyze the change in environmental conditions. The behaviour factor estimation using the federated learning process is depicted in Fig. 5.

The learning process for behaviour factor assessment is depicted in Fig. 5. This learning process inputs $\text{Len}_{\text{pat}} \forall t$ provided $\text{amt}_{\text{paid}} = R_{\text{rate}}$ is increasing over the tenure. Therefore, amt_{paid} is directly proportional to R_{rate} for t such that $\left(U_{\text{repay}} \propto \frac{1}{\text{amt}_{\text{due}}}\right)$ is true. This is the linear case for $B_{\text{fact}}(t)$ computation without $(Br_c - Dr_i)$ is used

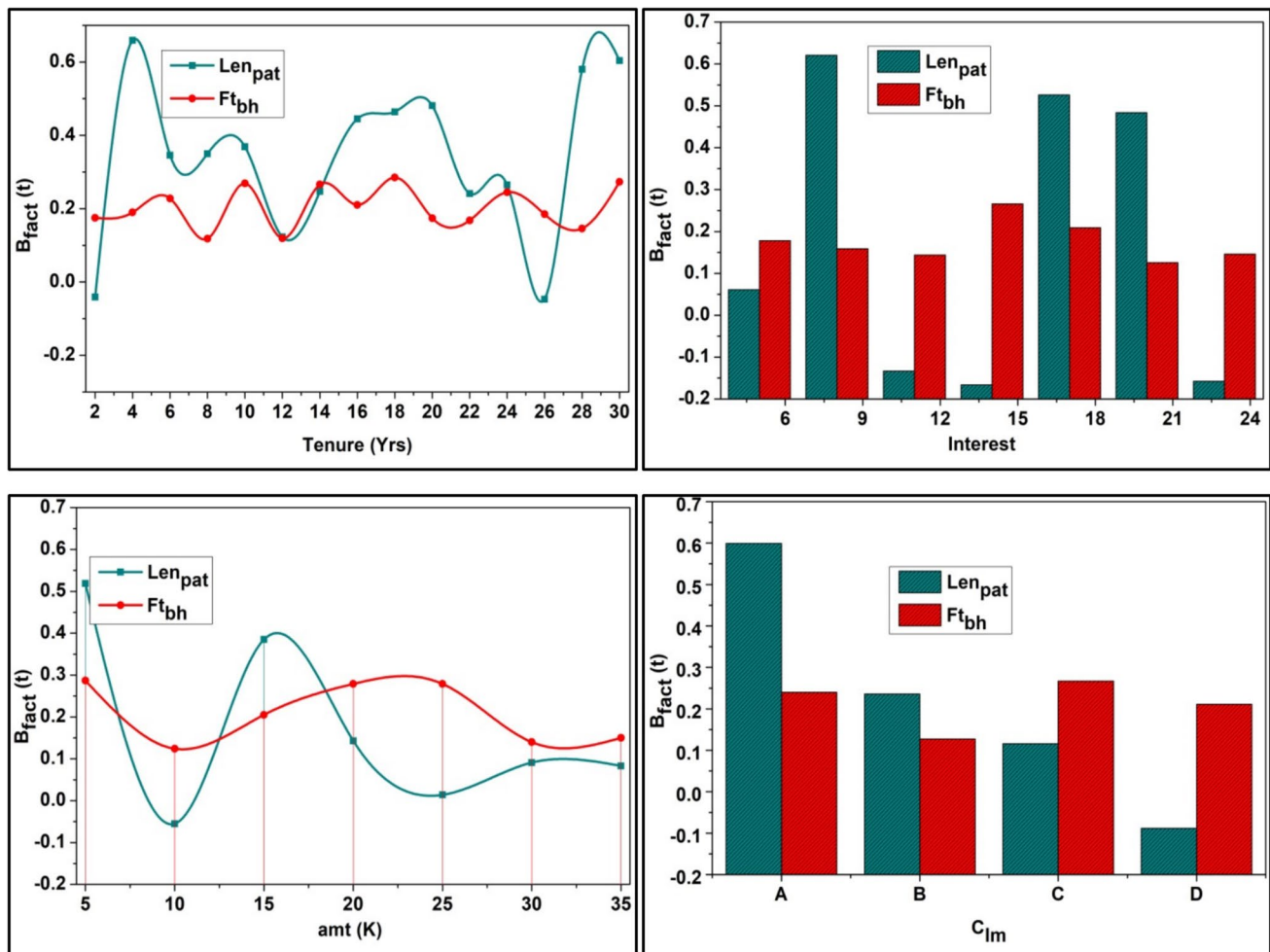


Fig. 6 Behaviour factor analysis

to identify multiple behavior patterns under R_{rate} and Ft_{bh-in} inputs. However, there is an alternate case for validating $f(C_{period})$ under $(Br_c + C_{hist}) \geq \psi$ and verifying $Dr_i \leq \psi$. This condition verification is used for $\max_t C_{period}$ and $\min_t C_{period}$ under B_{fact} . The combinations of (Br_c, C_{hist}) are used to detect $Ft_{bh}(t)$ for U_{replay} under new patterns detected. In Fig. 6, B_{fact} is an analysis of different variants.

In micro-lending, a borrower's behaviour over time is analyzed based on the credit period. The proposed method indicates a high behaviour factor value, reflecting positive financial behaviour. The repayment patterns with financial adaptability $Ft_{bh}(t) \forall U_{replay}$ were analyzed based on the economic conditions to avoid risk during lending. Federated learning ensures fairness and accuracy throughout the entire process. A low behaviour factor occurs when a borrower has missed payments and engages in irregular financial activity. A high behaviour factor occurs when a borrower timely repays their amount and consistent economic market. This ensures the borrower's financial health and adaptability, contributing to positive lending decisions (Fig. 6). The federated learning process is described in Algorithm 3.

Algorithm 3 Federated Learning Process

Read C_{lm}, Br, C_{risk} and I_{rate}

Verify $R_{rate} > C_{lm} \forall (amt - E_{con})$

Initialize $R_{amt-tot} = \sum_{i=1}^t \left(B_{fact,i} \times \left(\frac{C_{risk} \times I_{rate,i}}{1 - (1 + I_{rate,i})^t} \right) \right)$

Perform $R_{rate} = \left(\frac{R_{amt-tot}}{amt} \right) t$

Verify $R_{amt-tot}$

Generate $B_{fact,i}$ and C_{lm} to monitor $(1 + I_{rate,i})^t$

Compute $C_{lm} = [B_{fact} \times Br_{risk}] + (1 + C_{hist}) \times \left(\frac{C_{risk}}{E_{con}} \right)$

Monitor $C_{lm} \propto \left(\frac{C_{risk}}{E_{con}} \right)$

If $0 \leq Br_{risk} \leq 1$

Validate $Br_{risk} = \sum_{i=1}^t (amt + (Br_c - C_{hist}))$

If $Br_{risk} > 1$

Verify $R_{rate}(U_{repay})$ and $C_{stable}(U_{repay})$ to identify risk under different re-payment tenures

Compute $H_{res}(t) \rightarrow (R_{rate}(U_{repay}) + C_{stable}(U_{repay}))$

Ensure $Len_{pat} = \sum_{i=1}^t \left(\frac{amt \times C_{risk}}{1 - (1 + I_{rate})} \right) + (1 - Dr_i)$ to measure the process of lending for credit risk.

Perform $Len_{fed}(t) = \sum_{i=1}^t (Len_{pat} \times C_{risk}(t + 1)) \times \left(\frac{C_{stable}(U_{repay}) + Br_c}{(1 + I_{rate})} \right)$

This helps to measure the borrower's factors even during high default rates and liabilities in the lending process. The risk filtering is expressed as A_{risk} in the following equation.

$$A_{risk} = \sum_{i=1}^t \left(\frac{C_{risk} \times C_{stable}}{I_{rate} - \max_t C_{period}} \right) - \left(\frac{C_{risk} \times U_{repay}}{I_{rate} - \min_t C_{period}} \right) + (R_{rate} \times C_{lm}) + (Dr_i \times E_{con}). \quad (13)$$

Here, the term $\left(\frac{C_{risk} \times C_{stable}}{I_{rate} - \max_t C_{period}} \right)$ identifies the maximum risk factors based on the maximum credit period with credit stability. The term $\left(\frac{C_{risk} \times U_{repay}}{I_{rate} - \min_t C_{period}} \right)$ measures the minimum risk factors based on the minimum credit period with repayment rate. The difference between maximum and minimum credit periods was analyzed to filter the risk factor. Non-risk factors highlighted the risk occurrence factors to obtain accurate risk factors in the lending

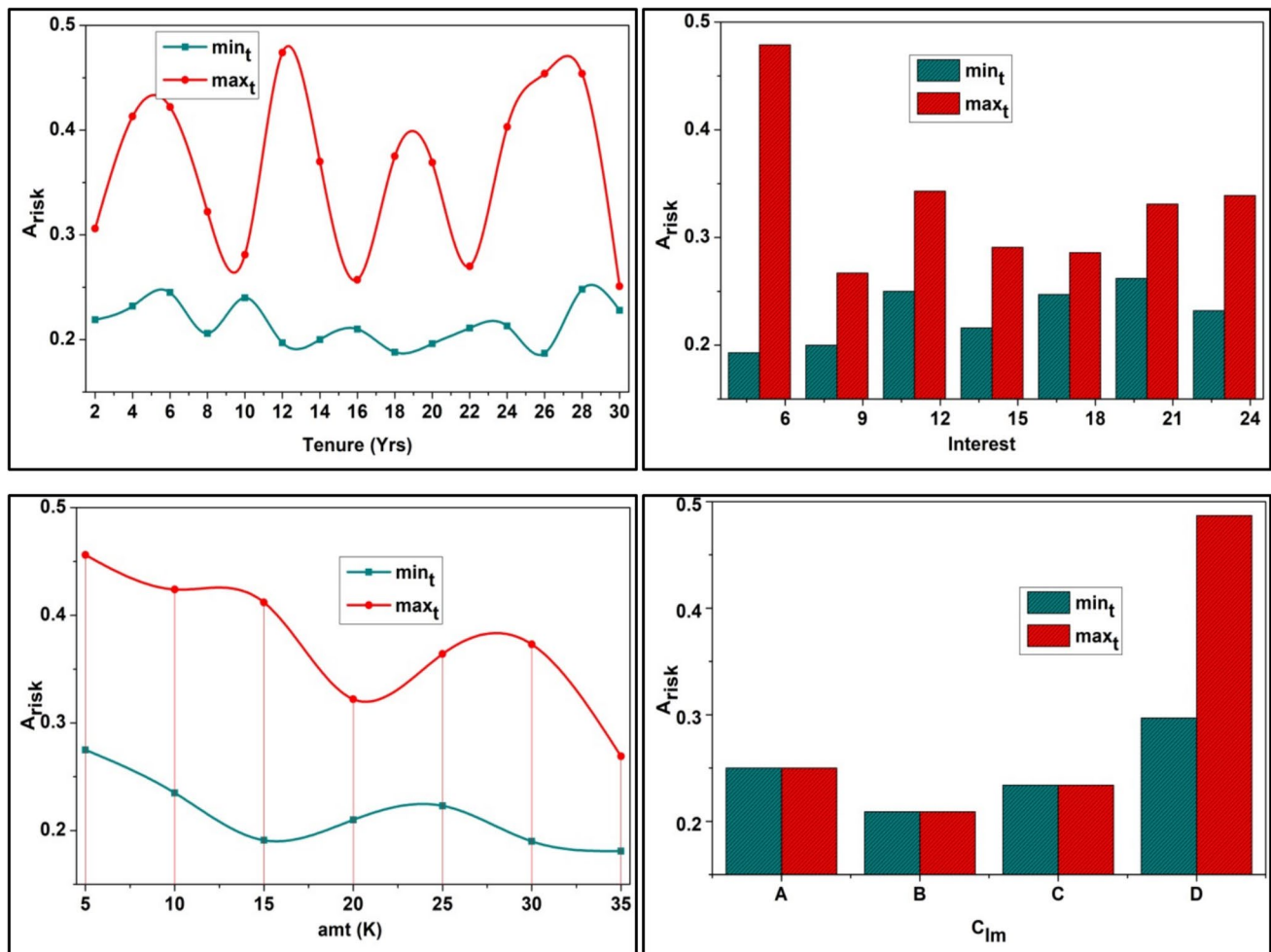


Fig. 7 Filtering ratio analysis

process. New credit risks are identified using this filtered factor from the previous return period. This filtering process helps to make decisions in the micro-lending environment. The following equation $risk_{detect}(t)$ detects the risk.

$$risk_{detect}(t) = (Br_C + C_{hist} + C_{lm}) \times \left(\frac{1}{A_{risk}} \right) \times (B_{fact} - C_{period}) + E_{con}. \quad (14)$$

The borrower credits with history and credit limit were analyzed to predict the external economic risk that was not evaluated during the lending process. The term $risk_{detect}(t) \propto \left(\frac{1}{A_{risk}} \right)$ indicates that an increase in accurate filtering decreases the risk. The term $(B_{fact} - C_{period})$ normalizes the behaviour pattern by differentiating the credit period to monitor the financial instability. When $risk_{detect}(t) > A_{risk}$ indicates high risk based on the risk filter process. This helps to address the borrower's ability to manage the risk and evaluate the lender to modify the borrowing amount during the micro-lending process. These identifications define new behavioural patterns for the selected credit limit. The filtering ratio is analyzed as presented in Fig. 7.

The proportion of risk factors to non-risk factors with borrowers was monitored to filter the risk. A high filtering ratio $A_{risk} = \sum_{i=1}^t \left(\frac{C_{risk} \times C_{stable}}{I_{rate} - \max_t C_{period}} \right) - \left(\frac{C_{risk} \times U_{repay}}{I_{rate} - \min_t C_{period}} \right)$ indicates a larger proportion of risks being detected

and filtered effectively by the proposed method. It ensures the effectiveness of the CFM in separating risk-prone borrowers from low-risk. For example, a borrower with a high filtering ratio and poor repayment behaviour will have high leverage in lending. Meanwhile, a borrower with consistent repayments and a low filtering ratio will have low leverage in lending. This proposed method continuously refines its analysis through federated learning to prioritize actionable risks from the obtained risk factors. A high filtering ratio enhances precision in isolating risky behaviours and ensures a robust assessment (Fig. 7). The risk detection process is explained in Algorithm 4.

Algorithm 4 Risk Detection Process

Generate Br_C, C_{lm} and C_{hist}

Compute $Ft_{bh}(t) = (Len_{fed}(t) \times Ft_{bh-in}) \forall U_{repay}$

Evaluate $C_{period} = \left(1 + \left(\frac{amt}{amt_{mon}}\right)\right) + \log(1 + I_{rate}) + \left(\frac{Dr_i}{E_{con}}\right)$

If $(Br_C + C_{hist}) \geq \varphi$ and $Dr_i \leq \varphi$

Then ensure $C_{period} \rightarrow \max_t C_{period}$

If $(Br_C + C_{hist}) < \varphi$ and $Dr_i > \varphi$

Then ensure $C_{period} \rightarrow \min_t C_{period}$

Analyze $C_{period} \forall U_{repay}$ to monitor $B_{fact}(t)$

Compute $U_{repay} = \left(\frac{amt_{due}}{amt_{paid}}\right) \times amt \times (R_{rate} \times C_{lm})$

Monitor $Len_{fed}(t) = \sum_{i=1}^t \left(Len_{pat} \times C_{risk}(t+1)\right) \times \left(\frac{C_{stable}(U_{repay}) + Br_C}{(1+I_{rate})}\right)$

Analyze $B_{fact}(t) = \sum_{i=1}^t \left(C_{period} \times Ft_{bh}(t)\right)$

Include O_i to measure $\min_t C_{period} \leq C_{period} \leq \max_t C_{period}$

Analyze $A_{risk} \forall C_{period}$ and $risk_{detect}(t) > A_{risk}$

Compute $A_{risk} = \sum_{i=1}^t \left(\frac{C_{risk} \times C_{stable}}{I_{rate} - \max_t C_{period}}\right) - \left(\frac{C_{risk} \times U_{repay}}{I_{rate} - \min_t C_{period}}\right)$

Evaluate $risk_{detect}(t) = (Br_C + C_{hist} + C_{lm}) \times \left(\frac{1}{A_{risk}}\right) \times (B_{fact} - C_{period})$

The potential for bias in credit scoring algorithms may be reduced using several methods. Methods for pre-processing data, such as re-sampling (for instance, reweighting underrepresented groups) and SMOTE (for example, synthetic data production), may help achieve class distribution parity. Model training may be adjusted to eliminate inequalities using in-processing approaches such as adversarial debiasing or fairness-aware loss functions. Adjustments to thresholds and re-ranking procedures are examples of post-processing solutions that guarantee fairness in final decision-making without changing the underlying model. Decentralized fairness-aware optimization approaches and individualized model changes are necessary to minimize systemic biases, particularly in federated learning contexts.

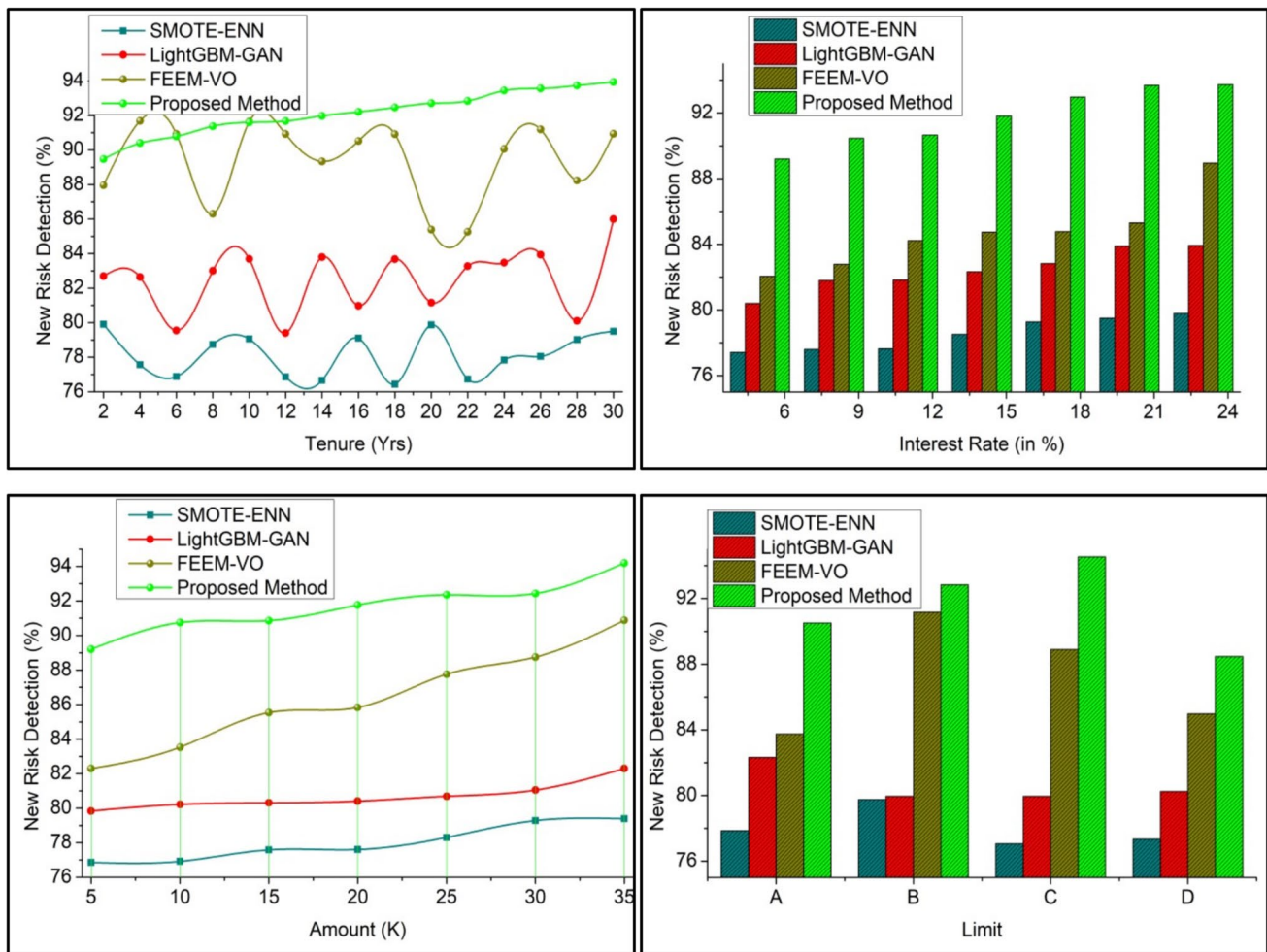


Fig. 8 New risk detection

5 Results and Discussion

The results for two new risk and return rate metrics are discussed in this part. The results are analyzed using the WEKA tool by analyzing the raw input data formatted for credit risk analysis. In this results part, the existing SMOTE-ENN (referenced in [21]), Light GBM-GAN (referenced in [20]), and FEEM-VO (Referenced in [22]) methods are considered for comparative analysis. The X-variants are: tenure (2 to 20 yrs.), interest rate (5 to 23), amt(K) (5 to 35), and C_{lm} (A to D) considered. The comparative analysis results are graphically presented in Figs. 8 and 9 for the above metrics.

To enhance the reliability of the findings, this research incorporates 95% confidence intervals (for instance, $\pm 2.1\%$ for accuracy) and statistical significance tests such as a paired t-test ($p < 0.05$) and Wilcoxon signed-rank test. For reliability, information on hyperparameter adjustment is additionally required. Applying Bayesian Optimization across a fivefold cross-validation setup, the following parameters were optimized for the XGBoost model used for credit risk assessment: number of boosting rounds = 500, learning_rate = 0.05, max_depth = 6, subsample = 0.8, fraction of features used per tree = 0.7, and L2 regularization lambda = 1.2.

One possible source of bias in the dataset is the difference between Group A (the majority) and Group B (the minority). Group A has an average credit score of 720, an acceptance rate of 85% for loans, and a default rate of 5%. In contrast, Group B has a lower approval rate of 65%, an average score of 680, and a slightly higher default rate of 7%. One way to address these biases is by using fairness-aware techniques. These techniques

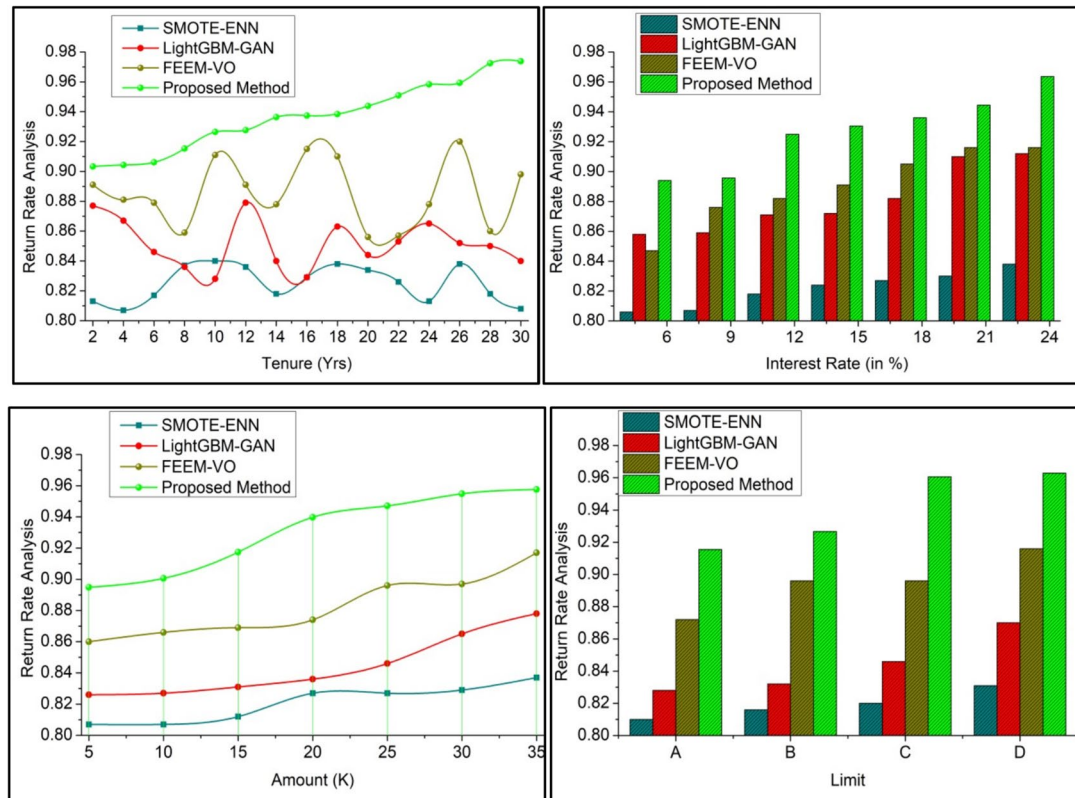


Fig. 9 Return rate analysis

include reweighting training data to ensure underrepresented groups are balanced, using equalized odds constraints to ensure risk-equivalent applicants have consistent approval rates, and using explainability methods like SHAP or LIME to find biased feature contributions and adjust them. Table 2 shows the experimental setup.

The occurrence of newly identified risks with time-based on borrower specification is analyzed using the proposed method. The proposed method ensures fewer values, which indicates a minimal emergence of new risks. Federated learning continuously updates the models based on previously unseen risks identified based on credit history. The accurate filtering and pattern analysis reduce the emergence of new risks and ensure the stability of the credit assessment in micro-lending. If a borrower with variation in financial behaviour stabilizes their repayment patterns, there will be a high possibility of new risk. A borrower with predictable and stable financial behaviour will have a low occurrence of new risk (Fig. 8).

The return rate in micro-lending represents the proportion of borrowers repaying loans on time. The proposed CFM with LP analysis improves the return rate compared to existing methods. The existing methods often fail to adapt borrower behaviours based on economic conditions. It leads to lower and unstable return rates during economic downturns. The proposed method continuously updates borrower credit patterns using federated learning to monitor repayment behaviours during unstable economic conditions and $R_{\text{rate}} > C_{\text{lm}} \forall (\text{amt} - E_{\text{con}})$ indicates the possibility of risk. This helps to identify where the return rate declines during market fluctuations and environmental factors. It maintains stability to enhance risk filtering and improves repayment outcomes in a micro-lending environment (Fig. 9). Table 3 shows the comparative analysis.

As data distributions vary over time, a phenomenon known as model drift occurs, and the prediction performance continues to decline. The problem is much more severe in federated networks when clients' data is distributed differently, making global model updates less effective. Adaptive federated optimization methods like FedAvg with momentum, FedProx (which regularizes local model updates), and drift-aware aggregation procedures are

Table 2 Experimental setup

Component	Details
GPU	NVIDIA A100 (80 GB VRAM)
CPU	64-core AMD EPYC 7742
RAM	128 GB
Storage	2T B
Network	10 Gbps Ethernet
Operating system	Windows 11
Programming language	Python 3.9
Bias mitigation	AIF360, Fairlearn
Learning rate	0.01
Max depth	10
Number of estimators	500

Table 3 Comparative analysis

Model	New risk detection (%)	Return rate analysis (%)	Tenure (yrs) (X-axis)	Interest rate (%)	Amount (K)	Limit categories (A, B, C, D)
SMOTE-ENN [21]	80.12	0.81	10	12	15	A: 78, B: 80, C: 82, D: 84
LightGBM-GAN [20]	86.45	0.88	15	15	20	A: 82, B: 85, C: 88, D: 90
FEEM-VO [22]	90.32	0.92	20	18	25	A: 86, B: 89, C: 91, D: 93
CFM-LPA (proposed)	95.21	0.97	25	21	30	A: 90, B: 93, C: 95, D: 96

necessary to combat drift. Concept drift detection approaches, such as adaptive windowing (ADWIN) or Kolmogorov–Smirnov tests, may be used to monitor model performance continuously. This allows for timely model upgrades to be triggered to retain accuracy. One adversarial attack is model poisoning, in which malicious clients insert biased gradients into a federated learning system. Another type is inference attacks, in which attackers try to retrieve private data from shared model updates. Defending against poisoning attacks may be done via defensive measures like Byzantine-resilient aggregation (like Krum or median-based filtering). To avoid inference attacks without compromising model usefulness, utilize differential privacy strategies like secure multi-party computing (MPC) or local perturbation (e.g., Gaussian noise addition). Further improvements to federated learning framework security and trust may be achieved via robust encryption methods and consensus mechanisms based on blockchain technology.

6 Conclusion

The proposed CFM with LP analysis is transformative in credit risk assessment for micro-lending. The method integrates federated learning and behavioural analysis to understand borrowers' repayment trends and credit stability. The continuous evaluation of return rates with credit limits and consumer responses predicts the risk assessment with the financial behaviours of the borrowers. This approach improves the accuracy of credit risk models that protect data. The filtering and utilization of behavioural factors from previous return periods allow micro-lending organizations to adjust their approach according to the dynamics of the borrowers. It helps to avoid risks and enhance sustained growth in the micro-lending platform. The proposed method reduces default rates and ensures stronger financial systems with high-risk detection in the micro-lending. The proposed method improves the risk detection analysis by 14.82% and return rate analysis by 13.63% for the different interest rates. However, the model's reliance on collaborative filtering may introduce biases if the available lending data is

sparse or imbalanced, potentially leading to inaccurate risk predictions for new or underrepresented borrowers. Federated learning ensures data privacy; it requires significant computational resources and robust synchronization across multiple lending institutions, which may pose scalability challenges. Future studies will improve the model's scalability through efficient communication protocols in federated learning, help reduce computational overhead and incorporate real-time financial indicators and alternative credit data sources, such as social and transactional behaviour, to improve adaptability to dynamic lending environments.

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Data Availability Data is open accessible through the link: <https://www.kaggle.com/datasets/laotse/credit-risk-dataset>.

Declarations

Conflict of Interest The authors declare no competing interests.

Informed Consent Not applicable.

Institutional Review Board Statement Not applicable.

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References

1. Aiello, M.A., Angelico, C.: Climate change and credit risk: the effect of carbon tax on Italian banks' business loan default rates. *J. Policy Model.* **45**(1), 187–201 (2023)
2. Dömötör, B., Illés, F., Ölvedi, T.: Peer-to-peer lending: legal loan sharking or altruistic investment? analyzing platform investments from a credit risk perspective. *J. Int. Financ. Markets. Inst. Money* **86**, 101801 (2023)
3. Kaya, O.: The impact of late payments on SMEs' access to finance: evidence from credit rationing and loan terms. *Econ. Model.* **141**, 106896 (2024)
4. Chan, S.H.: An exploratory study of using micro-credit to encourage the setting up of small businesses in the rural sector of Malaysia. *Asian Bus. Manag.* **4**, 455–479 (2005)
5. Enimu, S., Eyo, E.O., Ajah, E.A.: Determinants of loan repayment among agricultural microcredit finance group members in Delta state, Nigeria. *Financ. Innov.* **3**, 1–12 (2017)
6. Dorfleitner, G., Oswald, E.M., Zhang, R.: From credit risk to social impact: on the funding determinants in interest-free peer-to-peer lending. *J. Bus. Ethics* **170**, 375–400 (2021)

7. Daniels, K., Ramirez, G.G.: Information, credit risk, lender specialization and loan pricing: evidence from the DIP financing market. *J. Financ. Serv. Res.* **34**, 35–59 (2008)
8. Meressa, H.A.: Micro-and small-scale enterprises' financing preference in line with POH and access to credit: empirical evidence from entrepreneurs in Ethiopia. *J. Innov. Entrepreneurship* **11**(1), 54 (2022)
9. Shrieves, R.E., Dahl, D.: Regulation, recession, and bank lending behavior: the 1990 credit crunch. *J. Financ. Serv. Res.* **9**(1), 5–30 (1995)
10. Blanco-Oliver, A., Samaniego, A., Palacin-Sanchez, M.J.: How do loan officer-borrower gender-driven behavioural differences impact on the microfinance lending market? *Borsa Istanbul. Rev.* **24**(3), 435–448 (2024)
11. Hu, B., Hu, Y.P.: Pricing models for small and micro loan portfolio insurance. *Int. Rev. Financ. Anal. Financ. Anal.* **96**, 103552 (2024)
12. Chu, L., Mathieu, R., Robb, S., Zhang, P.: Bank capitalization and lending behavior after the introduction of the basle accord. *Rev. Quant. Financ. Acc. Financ. Acc.* **28**, 147–162 (2007)
13. Andersson, P.: P1198: Software for tracing decision behavior in lending to small businesses. *Behav. Res. Methods Instrum. Comput. Res. Methods Instrum. Comput.* **33**, 234–242 (2001)
14. Barboza, G., Trejos, S.: Micro credit in Chiapas, México: poverty reduction through group lending. *J. Bus. Ethics* **88**, 283–299 (2009)
15. Yang, F., Ye, X., Huang, W., Zhao, X.: The impacts on informal financing strategy of small and micro enterprises by interest rate risks and public health emergencies. *Int. Entrepreneurship Manag. J.* **19**(4), 1673–1705 (2023)
16. Wu, S., Dong, M., Tan, S., Dong, Y.: Who is lending to small and micro family business in China: evidence from CHFS data. *Small Bus. Econ.* **63**(3), 1225–1247 (2024)
17. Zhao, Y.: Investigation of the application of machine learning algorithms in credit risk assessment of medium and micro enterprises. *IEEE Access* (2024). <https://doi.org/10.1109/ACCESS.2024.3477556>
18. Liu, Y., Baals, L.J., Osterrieder, J., Hadji-Misheva, B.: Leveraging network topology for credit risk assessment in P2P lending: a comparative study under the lens of machine learning. *Expert Syst. Appl.* **252**, 124100 (2024)
19. Qian, Y., Wang, F., Zhang, M., Zhong, N.: Political uncertainty, bank loans, and corporate behavior: new investigation with machine learning. *Pac. Basin Financ. J. Financ. J.* **87**, 102480 (2024)
20. Zhuang, Y., Wei, H.: Design of a personal credit risk prediction model and legal prevention of financial risks. *IEEE Access* (2024). <https://doi.org/10.1109/ACCESS.2024.3466192>
21. Aruleba, I., Sun, Y.: Effective credit risk prediction using ensemble classifiers with model explanation. *IEEE Access* (2024). <https://doi.org/10.1109/ACCESS.2024.3445308>
22. Yang, D., Xiao, B.: Feature enhanced ensemble modeling with voting optimization for credit risk assessment. *IEEE Access* (2024). <https://doi.org/10.1109/ACCESS.2024.3445499>
23. Gamba-Santamaria, S., Melo-Velandia, L.F., Orozco-Vanegas, C.: Decomposition of non-performing loans dynamics into a debt-servicing capacity and a risk taking indicators. *Q. Rev. Econ. Finance* **96**, 101860 (2024)
24. Haitao, S.: Big data analysis of e-commerce loan risk of college students in the context of network finance. *IseB* **18**(3), 439–454 (2020)
25. Wang, F., Ding, L., Yu, H., Zhao, Y.: Big data analytics on enterprise credit risk evaluation of e-Business platform. *IseB* **18**(3), 311–350 (2020)
26. Rao, C., Liu, Y., Goh, M.: Credit risk assessment mechanism of personal auto loan based on PSO-XGBoost model. *Complex Intell. Syst.* **9**(2), 1391–1414 (2023)
27. Shetabi, M.: Evolutionary-based ensemble feature selection technique for dynamic application-specific credit risk optimization in FinTech lending. *Ann. Oper. Res. Oper. Res.* (2024). <https://doi.org/10.1007/s10479-024-06369-8>
28. Xia, H., Liu, J., Zhang, Z.J.: Identifying Fintech risk through machine learning: analyzing the Q&A text of an online loan investment platform. *Ann. Oper. Res. Oper. Res.* (2024). <https://doi.org/10.1007/s10479-020-03842-y>
29. Sinkey, J.F., Jr., Greenawalt, M.B.: Loan-loss experience and risk-taking behavior at large commercial banks. *J. Financ. Serv. Res.* **5**(1), 43–59 (1991)
30. Li, Z., Liang, S., Pan, X., Pang, M.: Credit risk prediction based on loan profit: evidence from Chinese SMEs. *Res. Int. Bus. Financ. Financ.* **67**, 102155 (2024)
31. Wang, Y., Zhang, Y., Liang, M., Yuan, R., Feng, J., Wu, J.: National student loans default risk prediction: a heterogeneous ensemble learning approach and the SHAP method. *Comput. Educ. Artif. Intell.* **5**, 100166 (2023)
32. Zhang, R., Lin, C., Tong, Z.: A visual risk identification and early warning research for college net loan based on microblog texts. *Risk Manage.* **23**, 261–281 (2021)
33. Liu, S., Xu, J.: Enterprise risk management, risk-taking, and macroeconomic implications: evidence from bank mortgage loan management. *J. Financ. Serv. Res.* (2024). <https://doi.org/10.1007/s10693-024-00422-0>
34. Carannante, M., D'amato, V., Fersini, P., Forte, S.: Machine learning-based climate risk sharing for an insured loan in the tourism industry. *Qual. Quant.* (2024). <https://doi.org/10.1007/s11135-024-01958-y>
35. Liu, P., Shao, Y.: Small business loan securitization and interstate risk sharing. *Small Bus. Econ.* **41**, 449–460 (2013)
36. <https://www.kaggle.com/datasets/laotse/credit-risk-dataset>

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Authors and Affiliations

Asma Aldrees¹ · Sana Shahab² · Ashit Kumar Dutta³ · Waseem Ahmad⁴ · Mohd Anjum⁵

✉ Waseem Ahmad
waseemahmad2@zhcet.ac.in

Asma Aldrees
edrees@kku.edu.sa

Sana Shahab
sshahab@pnu.edu.sa

Ashit Kumar Dutta
adotta@um.edu.sa

Mohd Anjum
mohdanjum@zhcet.ac.in

¹ Department of Informatics and Computer Systems, College of Computer Science, King Khalid University, Abha 61421, Saudi Arabia

² Department of Business Administration, College of Business Administration, Princess Nourah Bint Abdulrahman University, PO Box 84428, Riyadh 11671, Saudi Arabia

³ Department of Computer Science and Information Systems, College of Applied Sciences, AlMaarefa University, Ad Diriyah, Riyadh 13713, Kingdom of Saudi Arabia

⁴ Department of Computer Science and Engineering, Vishveshwarya Group of Institutions (VGI), Greater Noida, Gautam Buddha Nagar, Uttar Pradesh 201314, India

⁵ Department of Computer Engineering, Aligarh Muslim University, Aligarh 202002, India